

SOIL SPECTRA

The Soil Spectra is a **high-performance multi-parameter soil sensor** designed for **real-time monitoring** of essential soil properties. The device supports **RS485 Modbus RTU** for long-distance and highly reliable communication in agriculture and environmental monitoring applications. The sensor enables **precision farming** by capturing vital soil health parameters including **Moisture, Temperature, EC, pH, Nitrogen, Phosphorus, and Potassium**.

Key Features

- **7-in-1 Multi-parameter measurement**
(Moisture, Temperature, EC, pH, N, P, K)
- **Long-distance RS485 Modbus RTU** communication
- **Rugged IP65 enclosure** for continuous outdoor deployment
- **Fast response time < 1 sec**
- **High-accuracy field data** suitable for real-world agri conditions
- **Low power requirement** — **5V DC** makes it suitable for IoT systems



Technical Specifications

Parameter	Specification
Power Supply	5V DC
Output Signal	RS485
Protection Level	IP65
Temperature Range	-45°C to +115°C
Temperature Precision	±0.5°C
Moisture Range	0 – 100% RH
Moisture Precision	±3% (0-53%), ±5%(53-100%)
pH Measurement Range	0 – 14 pH
pH Precision	±0.3 pH
NPK Measurement Range	0 – 1999 mg/kg (each nutrient)

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Parameter	Specification
NPK Precision	±2% F.S.
EC Range	0 – 10000 µS/cm
EC Resolution	10 µS/cm
Response Time	< 1 second
Operating Environment	-20°C to 85°C, 0 – 100% RH

Enclosure & Mechanical Specifications

Characteristic	Specification
Body Material	Weather-resistant ABS polymer
Protection Rating	IP65 environmental protection
Color	Black
Maintenance Access	Quick-release fasteners for internal component access

Applications

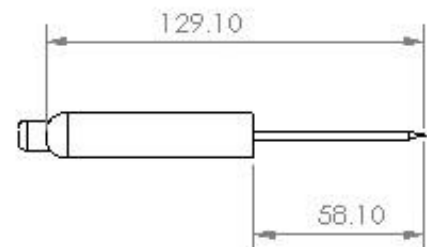
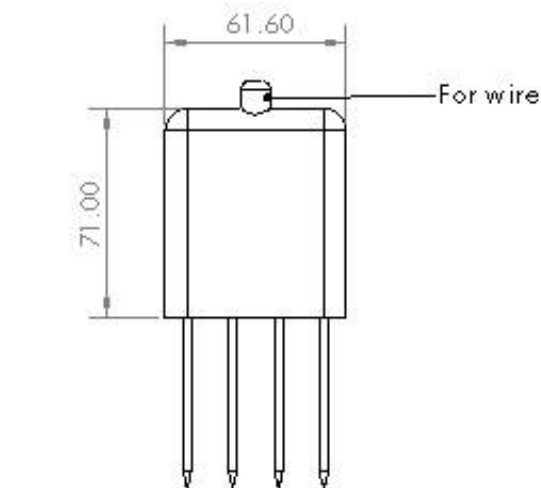
- Smart Agriculture & Irrigation Control:**Leverages real-time data on moisture, EC, and nutrients to optimize irrigation schedules and reduce water and fertilizer usage.
- Greenhouse Monitoring:**Ensures stable soil conditions in controlled environments by tracking temperature, pH, and nutrient levels, enabling precise crop management.
- Environmental Research:**Provides accurate, field-grade soil data critical for analyzing ecological patterns, climate impact studies, and long-term environmental assessments.
- Soil Quality Assessment:**Measures essential soil parameters like NPK, pH, and EC to evaluate fertility and inform land management and treatment strategies.
- Agri-Tech Solutions for Farms of All Sizes:** Compact, cost-effective, and scalable — suitable for both smallholder farms and large-scale agri-enterprises aiming to implement data-driven practices.
- Soil Mapping for Precision Farming:** Supports spatial variability analysis by enabling detailed soil health mapping across different zones of a field for site-specific interventions.

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- **Government & NGO Programs on Sustainable Agriculture:** Ideal for deployment in rural development projects focused on sustainability, resource conservation, and productivity enhancement through technology.

Quality Assurance

- Factory calibrated with traceable multi-parameter soil standards
- Compliance with industrial RS485 Modbus RTU communication protocols
- Environmental testing for extreme temperature and humidity operating ranges
- Vibration, probe-stress, and mechanical shock resistance validation for field use
- Long-term corrosion resistance testing against soil minerals and fertilizers
- Waterproof performance verification ensuring IP65 durability in buried conditions



Soil Spectra Base (All The Dimensions are in mm)